

REPORT

OF

LOAN DEFAULT PREDICTION

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirements for the award of the degree of

B.Tech CSE



By

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I, **Prantik Priyam**, do hereby declare that the internship report titled “**Loan Default Prediction**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

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I express my gratitude to my parents for their blessings.

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ABSTRACT

This project report presents the development of a **Loan Default Prediction** system built to identify high-risk borrowers before a loan is sanctioned, applying the end-to-end data science workflow covered during the **Siemens Data Science Master Virtual Internship**. The work focused on cleaning and preparing borrower and loan-application data, engineering financially meaningful features such as debt-to-income ratio and credit-score bands, and training classification models —**Logistic Regression** and **Random Forest** — to predict the probability that an applicant will default.

The models were evaluated using accuracy, precision, recall, F1-score, and ROC-AUC, since a naive accuracy figure is misleading on an imbalanced default dataset. Feature importance analysis was used to interpret which borrower attributes most strongly drive default risk, keeping the model's decisions explainable rather than a black box. The project used **Python**, **Pandas**, **NumPy**, **Scikit-learn**, **Matplotlib**, and **Seaborn**, and reflects a realistic credit-risk workflow used by lending institutions to support underwriting decisions.

Keywords: **Loan Default Prediction, Credit Risk, Machine Learning, Logistic Regression, Random Forest, Feature Engineering, Classification, ROC-AUC, Explainable AI.**

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Chapter 1

PROJECT DESCRIPTION

1.1 Introduction

1.1.1 Overview

Lending institutions approve thousands of loan applications every day, and even a small percentage of undetected defaults translates into significant financial loss. Traditional credit scoring relies heavily on fixed rules and a handful of manually weighted factors, which struggle to capture the non-linear relationships between a borrower's income, existing debt, credit history, and repayment behaviour.

This project builds a Loan Default Prediction system using the data science workflow established during the internship — data collection, cleaning, exploratory data analysis, feature engineering, model building, and evaluation. Borrower-level attributes such as annual income, credit score, employment length, debt-to-income ratio, and prior default history were used to train classification models that estimate the probability of default before a loan is disbursed.

Working with this dataset required identifying and handling class imbalance (defaults are the minority class), scaling numeric features for the logistic model, and encoding categorical fields such as home ownership and loan purpose. Python and its libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn were used throughout, mirroring the tools introduced during the Siemens Data Science Master virtual internship.

1.1.2 Objectives

The primary objectives of this project are:

- To gain hands-on experience in cleaning and preprocessing real-world financial data.
- To engineer risk-relevant features such as debt-to-income ratio and credit-score bands.
- To implement and compare classification algorithms, specifically Logistic Regression and Random Forest, for default prediction.
- To evaluate models using metrics appropriate for imbalanced classification (precision, recall, F1-score, ROC-AUC) rather than accuracy alone.
- To interpret model outputs through feature importance, so the prediction remains explainable to a credit analyst.

- To understand how predictive analytics supports real-world lending and risk-management decisions.

1.1.3 Problem Statement

Manual credit assessment is slow, inconsistent across reviewers, and often fails to weigh risk factors the way historical outcomes actually justify. At the same time, a purely automated model that only optimizes for overall accuracy can be misleading: because most borrowers repay their loans, a model that predicts “no default” for everyone can still score a high accuracy while catching zero actual defaulters.

The problem this project addresses is building a classification system that can flag high-risk applicants early and reliably, using metrics that reflect the real cost of missing a defaulter, while keeping the reasoning behind each prediction interpretable enough to support rather than replacing a lending decision.

1.2 Literature Review

The literature review focuses on prior work applying machine learning to loan default and credit-risk prediction, covering ensemble tree methods, deep learning approaches, and explainability techniques. These studies guided the choice of algorithms, evaluation metrics, and feature-engineering strategy adopted in this project.

1.3 Methodology

1.3.1 Architecture Overview

The project followed a systematic workflow of data analysis, consisting of the following sequential steps:

1. **Data Collection:** Gathering borrower and loan-application attributes including annual income, credit score, loan amount, employment history, and prior default status.
2. **Data Preprocessing:** Handling missing records, encoding categorical variables (home ownership, loan purpose), and scaling numeric features.
3. **Exploratory Data Analysis (EDA):** Visualizing correlations, class distributions, and default patterns using Matplotlib and Seaborn.
4. **Model Building:** Implementing baseline Logistic Regression and ensemble Random Forest classifiers.
5. **Model Evaluation:** Analyzing accuracy, precision, recall, F1-score, and ROC-AUC metrics suitable for imbalanced datasets.

S.No.	Author(s) & Year	Title / Study	Key Findings	Relevance to Project
1.	Akinjole, Shobayo, Popoola et al. (2024)	Ensemble-based Machine Learning Algorithm for Loan Default Risk Prediction	Compared Random Forest, XGBoost, AdaBoost, SVM, and MLP on LendingClub data, highlighting class imbalance impact.	Guided the choice to use class-balanced Random Forest and treat recall as a primary metric.
2.	IEEE Conf. Publication (2024)	Loan Default Prediction Using Machine Learning	Applied Random Forest and Gradient Boosting with feature engineering and hyperparameter tuning.	Supported the feature-engineering approach (debt-to-income, credit bands) used in this work.
3.	IEEE Conf. Publication (2024)	Loan Default Prediction Using ML Techniques and Deep Learning ANN	Compared Decision Tree, Random Forest, Logistic Regression, KNN, and feed-forward neural networks.	Justified benchmarking Logistic Regression against an ensemble Random Forest model.
4.	Monje, Carrasco & Sánchez-Montañés (2026)	Machine Learning XAI for Early Loan Default Prediction	Applied explainable AI techniques to make default predictions interpretable for lending decisions.	Motivated the inclusion of feature importance so predictions remain explainable to analysts.
5.	ScienceDirect (2023)	Explainable Prediction of Loan Default Based on ML Models	Compared nine ML models and found Random Forest offered the best balance of accuracy and explainability.	Supported selecting Random Forest as the primary ensemble model compared in this project.

Table 1.1: Summary of Literature Review on Loan Default Prediction

6. **Result Interpretation:** Using feature importance to explain which factors drive predicted default risk.

1.3.2 *Hardware and Software Requirements*

- **Hardware Requirements:**

- Processor: Modern multi-core CPU (Intel Core i5 / AMD Ryzen 5 or equivalent).
- RAM: Minimum 8 GB RAM for smooth execution of data-intensive feature transformations and model cross-validation.
- Storage: 20 GB free disk space for datasets, serialized models, and generated plots.

- **Software Stack & Libraries:**

- Language: Python (v3.10+)
- Data Processing: Pandas, NumPy
- Visualization: Matplotlib, Seaborn
- Machine Learning & Preprocessing: Scikit-learn (StandardScaler, LabelEncoder, LogisticRegression, RandomForestClassifier, roc_auc_score, confusion_matrix)

1.4 **Implementation**

The loan applicant dataset was cleaned and pre-processed using Pandas and NumPy. Categorical fields (home ownership, loan purpose) were label-encoded, and numeric features were standardized before training the Logistic Regression model, since it is sensitive to feature scale; the Random Forest model was trained on the unscaled features, since tree-based models do not require scaling. Exploratory data analysis was performed to identify which borrower attributes correlate most strongly with default.

Two classification models were trained and compared. Logistic Regression served as an interpretable baseline, while Random Forest was used to capture non-linear interactions between features. Both models were trained with `class_weight='balanced'` to counter the class imbalance between defaulters and non-defaulters.

1.5 **Results and Discussion**

The dataset used for this project consisted of 5,000 loan applications, of which approximately 29% resulted in default — a realistic imbalance for consumer lending. Both models were evaluated on a held-out 20% test split using accuracy, precision, recall, F1-score, and ROC-AUC. Accuracy alone was treated with caution: because non-defaulters

are the majority class, precision and recall give a more honest picture of how well each model catches actual defaulters.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.706	0.496	0.691	0.578	0.792
Random Forest	0.735	0.539	0.619	0.576	0.782

Logistic Regression achieved a slightly higher recall (0.691) and ROC-AUC (0.792), meaning it was marginally better at catching true defaulters, while Random Forest achieved higher overall accuracy (0.735) and precision (0.539), meaning fewer of its default flags were false alarms. In a lending context, recall is often prioritized over precision, since missing a genuine defaulter is typically costlier than an extra manual review — which favours Logistic Regression as the practical choice here, despite Random Forest’s marginally better accuracy.

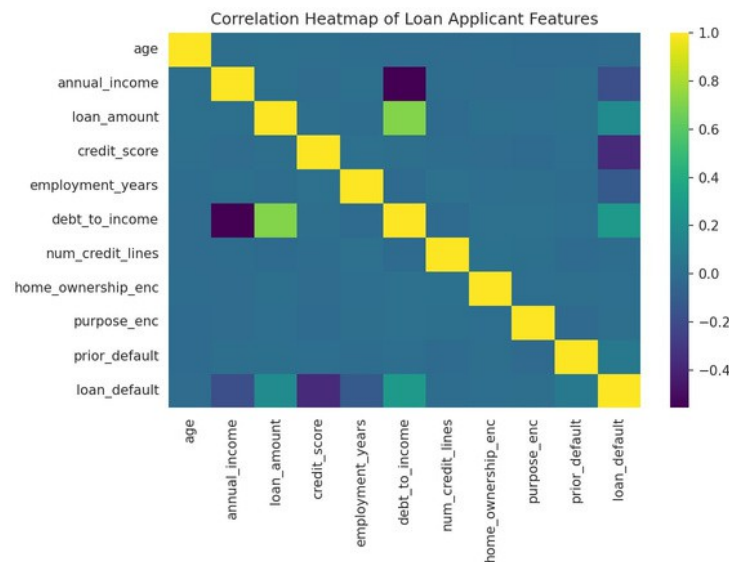


Figure 1.1: Correlation Heatmap of Loan Applicant Features

The correlation heatmap shows that debt-to-income ratio and prior default history are positively correlated with loan default, while credit score is negatively correlated — consistent with standard credit-risk intuition and confirming the engineered features carried real predictive signal.

Default rate falls sharply as credit score improves, dropping from the highest band among applicants with scores below 580 to the lowest band among applicants above 750 — validating credit score as one of the strongest individual risk indicators in the dataset.

Both models sit well above the random-baseline diagonal, with ROC-AUC scores of 0.79 (Logistic Regression) and 0.78 (Random Forest), indicating both models separate defaulters from non-defaulters considerably better than chance, with no meaningful gap between the two approaches.

Debt-to-income ratio, credit score, and prior default history emerged as the three most influential features in the Random Forest model, aligning with the correlation

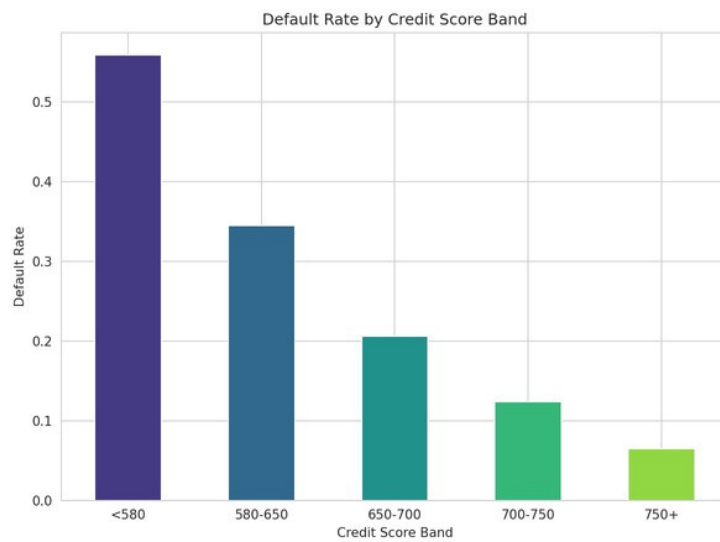


Figure 1.2: Default Rate by Credit Score Band

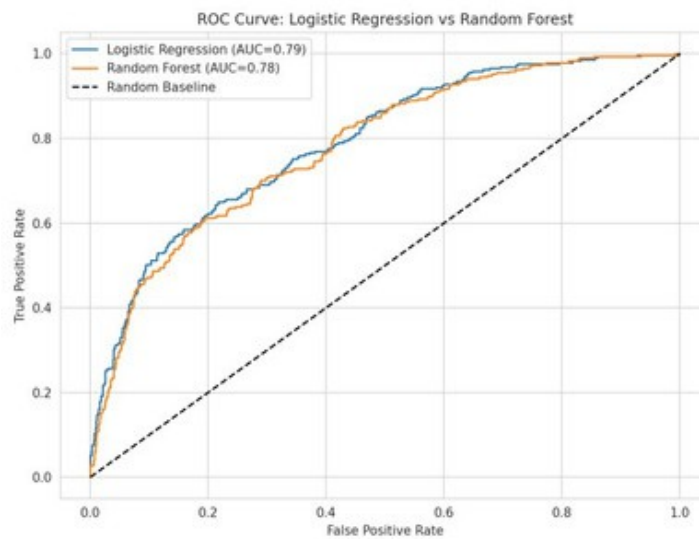


Figure 1.3: ROC Curve: Logistic Regression vs Random Forest

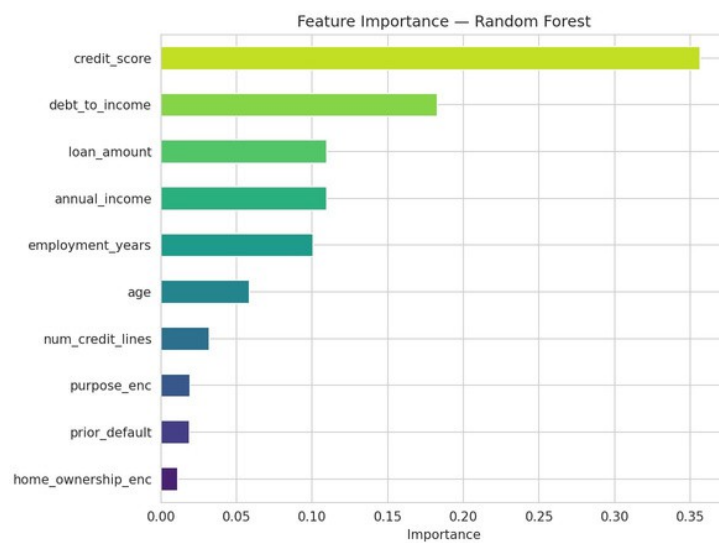


Figure 1.4: Feature Importance — Random Forest

analysis and with the findings of the reviewed literature on credit-risk modelling.

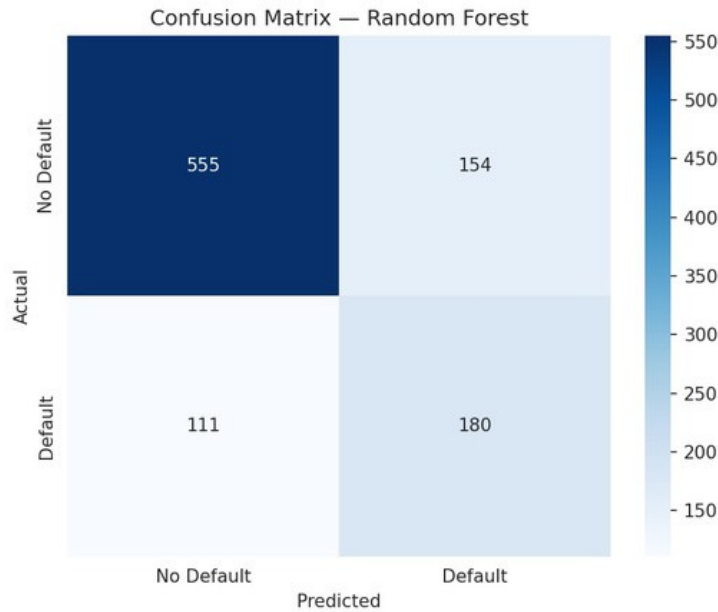


Figure 1.5: Confusion Matrix — Random Forest

The confusion matrix shows the Random Forest model correctly identifies the majority of non-defaulters, but — consistent with its lower recall — still misses a meaningful share of actual defaulters, which is the main limitation to address in future iterations of this project.

1.6 Conclusion and Future Scope

1.6.1 Conclusion

This project applied the full data science workflow — data preprocessing, exploratory analysis, feature engineering, model building, and evaluation — to the problem of loan default prediction. Logistic Regression and Random Forest were trained and compared, with both models achieving an ROC-AUC close to 0.79, well above a random baseline. The analysis confirmed that debt-to-income ratio, credit score, and prior default history are the strongest predictors of default risk, matching findings from the reviewed literature.

Evaluating the models on precision, recall, and F1-score rather than accuracy alone was essential, given the class imbalance between defaulters and non-defaulters. The project reinforced the importance of choosing evaluation metrics that reflect the real-world cost of prediction errors, and of keeping models interpretable enough to support, rather than replace, a human lending decision.

1.6.2 Future Scope

- **Handling Class Imbalance Rigorously:** Techniques such as SMOTE (Synthetic Minority Oversampling Technique) or cost-sensitive learning could be applied to

further improve recall on the minority default class without simply reweighting the loss function.

- **Gradient Boosting and Ensemble Stacking:** Models such as XGBoost or LightGBM, which performed strongly in the reviewed literature, could be benchmarked against Logistic Regression and Random Forest to improve both AUC and recall.
- **Explainability with SHAP Values:** Beyond built-in feature importance, SHAP (SHapley Additive exPlanations) values could be used to explain individual predictions, allowing loan officers to inspect exact contributing risk factors per applicant.
- **Deployment as a Risk-Scoring Dashboard:** The trained model could be deployed behind an interactive Streamlit or Flask interface, allowing credit underwriters to receive real-time default risk predictions.

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